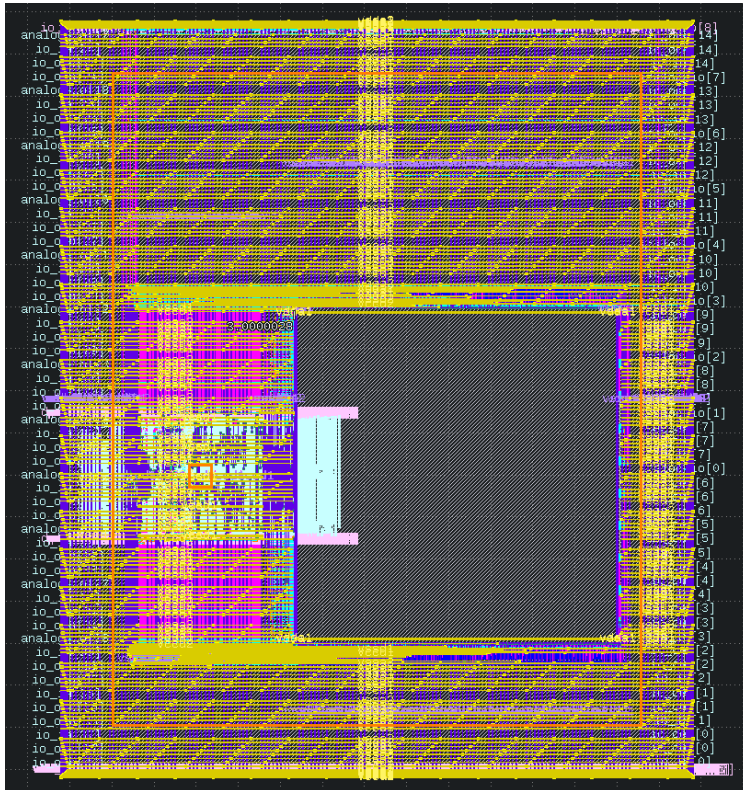


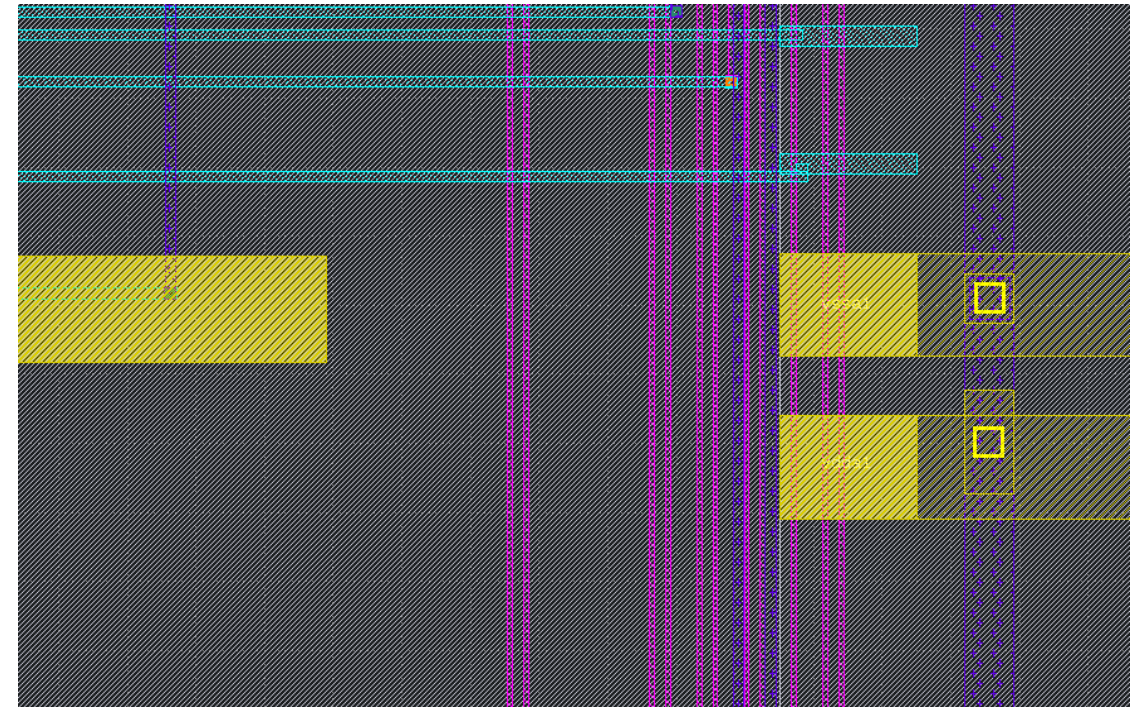
Issues w/PDN for Flattened Digital Design in Wrapper + Analog Macro

Both Macros – Digital + Analog

- Digital Block (left): VCCD1 & VSSD1
- Analog Block (right): VDDA1 & VSSA1

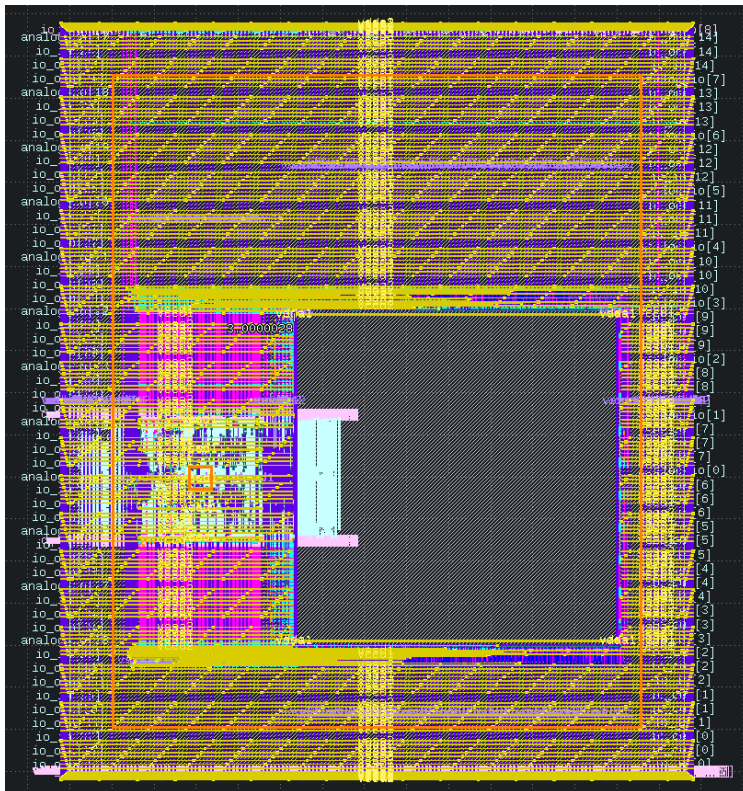


- Zoomed in top left corner of analog has a gap where power rails aren't connected to the grid (vccd1 and vssa1 here, but still no contacts to vdda1/vssa1)



Both Macros – Digital + Analog

- Digital Block (left): VCCD1 & VSSD1
- Analog Block (right): VD DA1 & VSSA1

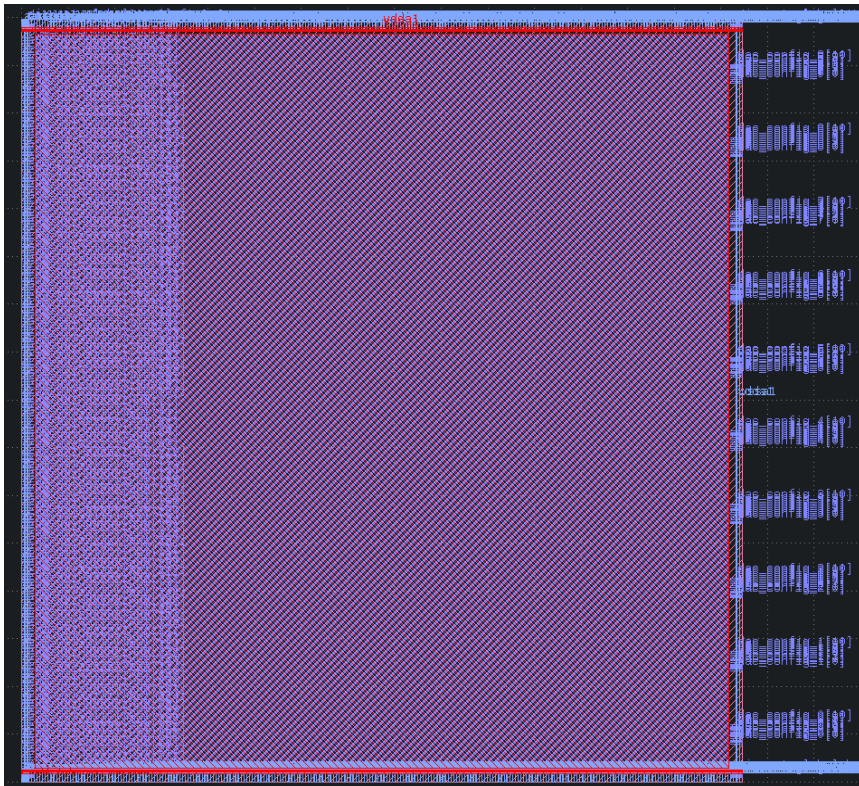


- This approach isn't favorable because we'll have long wires running to the edges of the wrapper
- Yielding large caps on long wires w/out buffering capability
- Source: [Integration & Power Routing](#) (options 1 & 2)

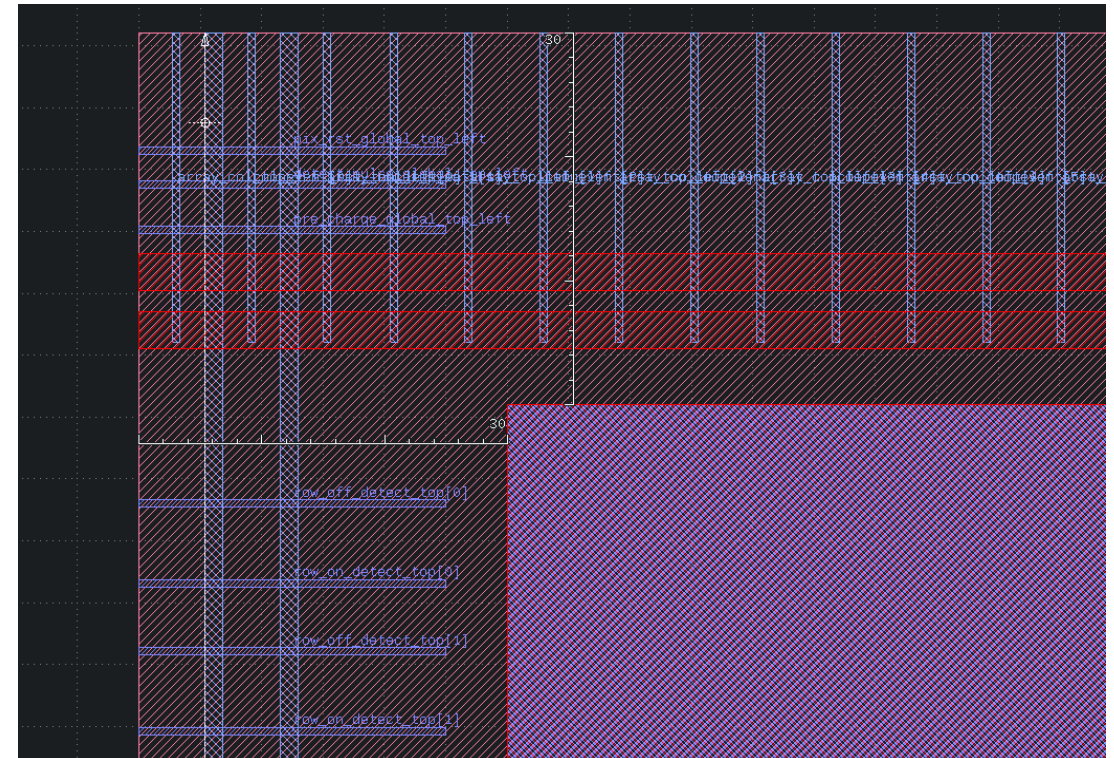
Corner/Group	Hold Worst Slack	Reg to Reg Paths	Hold TNS	Hold Vio Count	of which reg to reg	Setup Worst Slack	Reg to Reg Paths	Setup TNS	Setup Vio Count	of which reg to reg	Max Cap Violatio...	Max Slew Violati...
Overall	0.4180	0.4180	0.0000	0	0	0.8729	0.8729	0.0000	0	0	381	766
nom_tt_025C_1v80	0.7583	0.7583	0.0000	0	0	5.6031	5.6031	0.0000	0	0	276	555
nom_ss_100C_1v60	1.7671	1.7671	0.0000	0	0	0.9720	0.9720	0.0000	0	0	372	748
nom_ff_n40C_1v95	0.4189	0.4189	0.0000	0	0	7.3093	7.3093	0.0000	0	0	131	253
min_tt_025C_1v80	0.7569	0.7569	0.0000	0	0	5.6348	5.6348	0.0000	0	0	208	427
min_ss_100C_1v60	1.7638	1.7638	0.0000	0	0	1.1027	1.1027	0.0000	0	0	364	734
min_ff_n40C_1v95	0.4180	0.4180	0.0000	0	0	7.2930	7.2930	0.0000	0	0	117	242
max_tt_025C_1v80	0.7591	0.7591	0.0000	0	0	5.5884	5.5884	0.0000	0	0	287	583
max_ss_100C_1v60	1.7709	1.7709	0.0000	0	0	0.8729	0.8729	0.0000	0	0	381	766
max_ff_n40C_1v95	0.4187	0.4187	0.0000	0	0	7.3368	7.3368	0.0000	0	0	138	257

Analog Macro and Flattened Wrapper

- Digital Power: VCCD1 & VSSD1
- Analog Power: VDPA1 & VSSA1

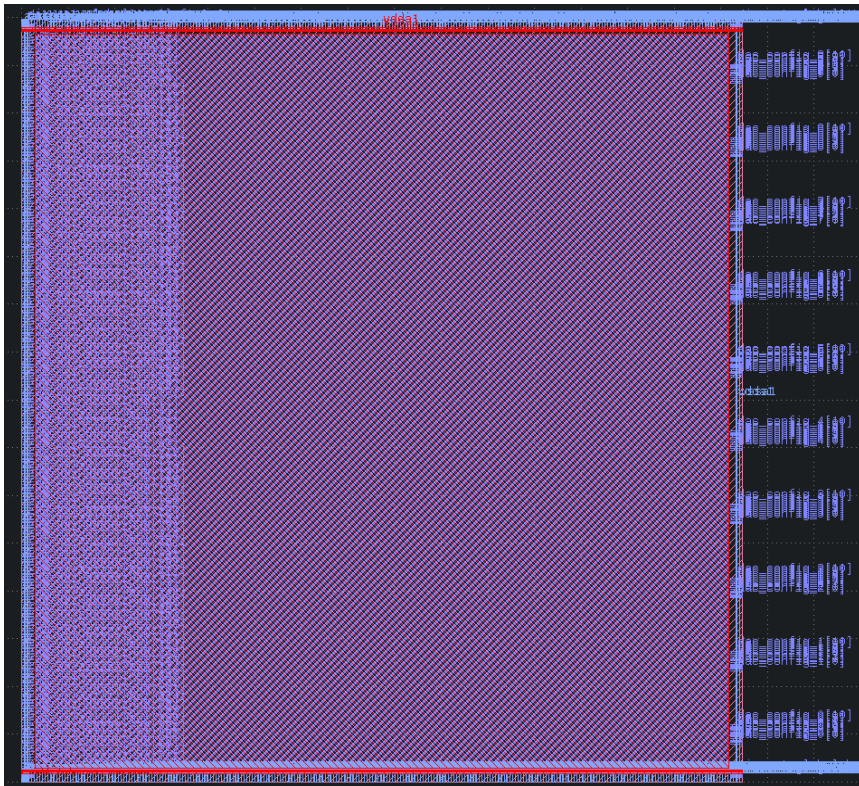


- Inserting a dummy LEF for the macro instance where VDPA1 & VSSA1 are on M4 & M5
- Zoomed in top left corner – 30um distance to OBS across all layers (no metal over photodiodes)



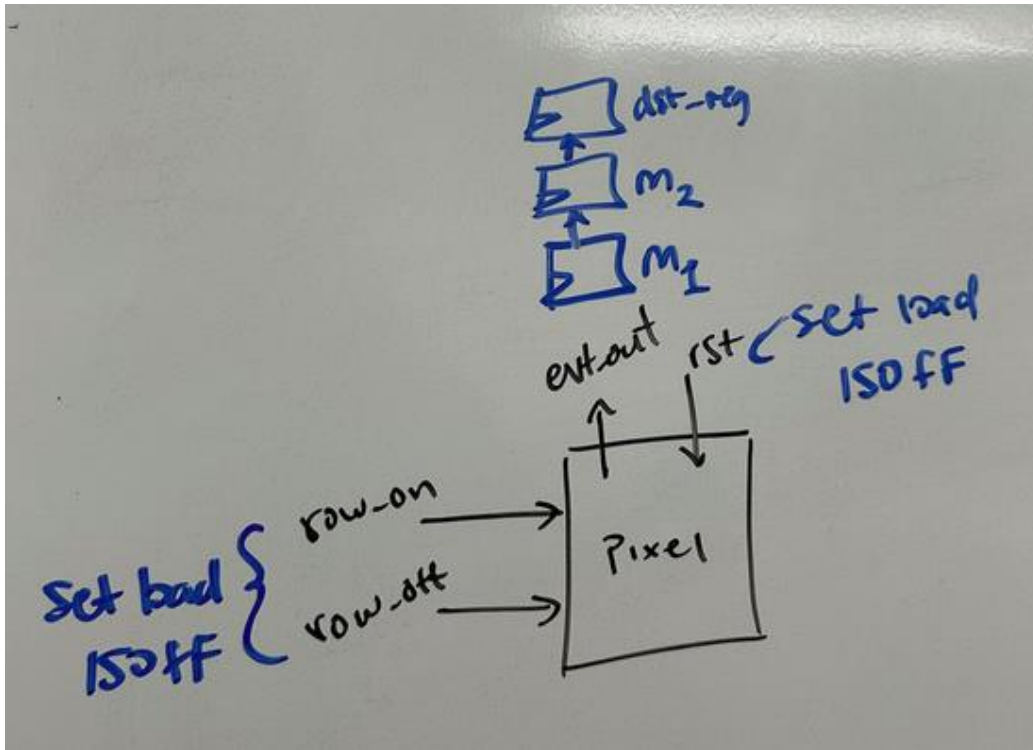
Analog Macro and Flattened Wrapper

- Digital Power: VCCD1 & VSSD1
- Analog Power: VDDA1 & VSSA1
- Running using similar parameters to what I had previously yield PDN warnings & errors
- PSM{0038,0039, **0069**,0210}, PDN{0110}



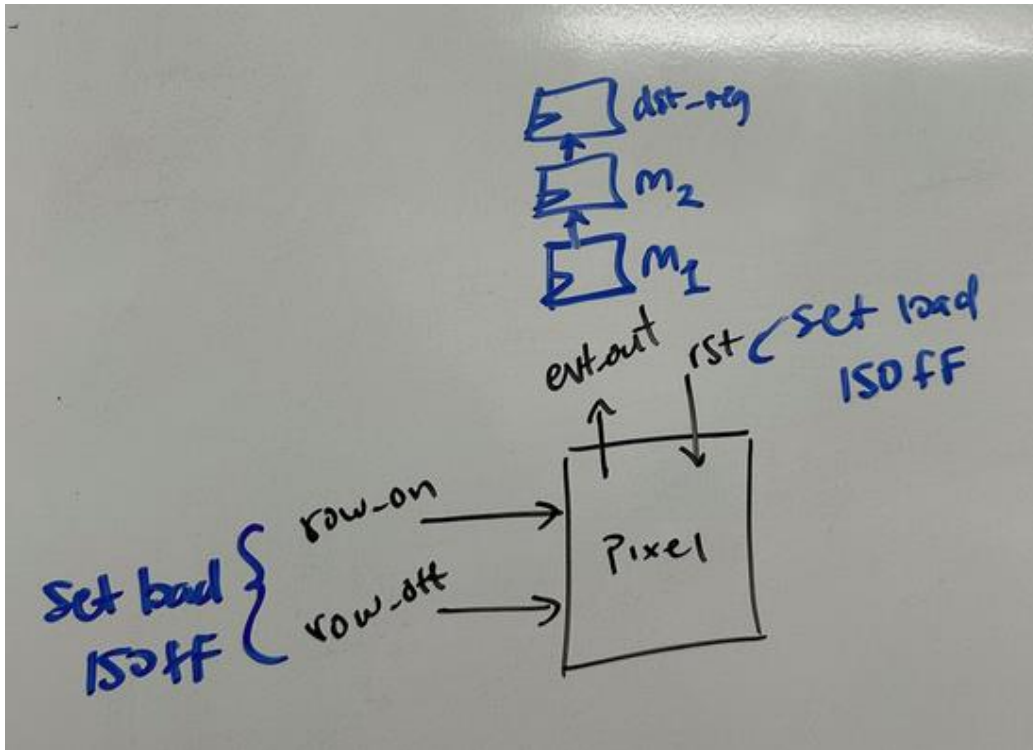
```
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_980_Left_2265/VPWR at location (6.210um, 2679.200um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_981_Left_2266/VPWR at location (6.210um, 2679.200um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_982_Left_2267/VPWR at location (6.210um, 2684.640um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_983_Left_2268/VPWR at location (6.210um, 2684.640um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_984_Left_2269/VPWR at location (6.210um, 2690.080um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_985_Left_2270/VPWR at location (6.210um, 2690.080um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_986_Left_2271/VPWR at location (6.210um, 2695.520um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_987_Left_2272/VPWR at location (6.210um, 2695.520um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_988_Left_2273/VPWR at location (6.210um, 2700.960um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_989_Left_2274/VPWR at location (6.210um, 2700.960um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_990_Left_2275/VPWR at location (6.210um, 2706.400um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_991_Left_2276/VPWR at location (6.210um, 2706.400um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_992_Left_2277/VPWR at location (6.210um, 2711.840um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_993_Left_2278/VPWR at location (6.210um, 2711.840um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_994_Left_2279/VPWR at location (6.210um, 2717.280um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_995_Left_2280/VPWR at location (6.210um, 2717.280um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_996_Left_2281/VPWR at location (6.210um, 2722.720um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_997_Left_2282/VPWR at location (6.210um, 2722.720um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_998_Left_2283/VPWR at location (6.210um, 2728.160um).
[05:30:36] WARNING [PSM-0039] Unconnected instance PHY_EDGE_ROW_999_Left_2284/VPWR at location (6.210um, 2728.160um).
[05:30:36] WARNING [PSM-0039] message limit reached, this message will no longer print
[05:30:36] ERROR [PSM-0069] Check connectivity failed on vccd1.
[05:30:36] WARNING Grid check for vccd1 failed: PSM-0069
[05:30:36] ERROR [PSM-0028] Cannot find net vccd2 in the design.
[05:30:36] WARNING Grid check for vccd2 failed: PSM-0028
[INFO PSM-0040] All shapes on net vdda1 are connected.
[05:30:36] ERROR [PSM-0028] Cannot find net vdda2 in the design.
[05:30:36] WARNING Grid check for vdda2 failed: PSM-0028
Classic - Stage 20 - Generate PDN 19/78 0:02:12
```

Parameters for Current Timing Constraints



- Digital Signals driving unknown analog load --> large buffering
- Analog data coming into digital domain --> metastability registers
- The current parameters force these components to be near the ports of the analog design

Parameters for Current Timing Constraints

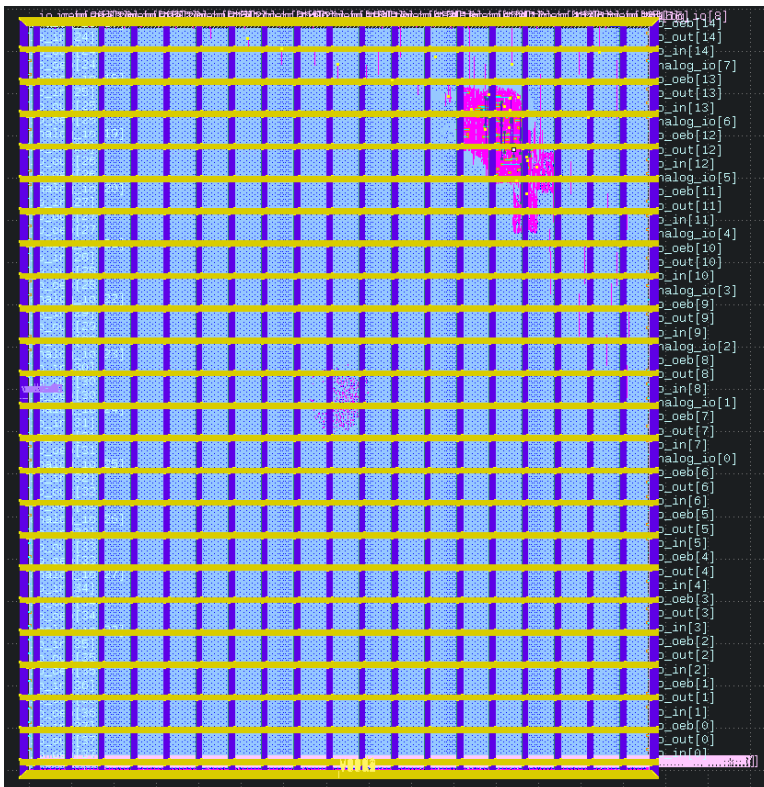


- A successful top level run:
 - PDN connection for digital VCCD1/VSSD1 and analog VDPA1/VSSA1
 - Buffers close to the input ports
 - Metastability registers grouped close to one another near array

Flattened Wrapper w/ Digital Only

- All nets are connected in this run
- Running the flow with just the verilog files results in a run that passes the PDN errors with a DRC/LVS/STA clean

- Digital Power: VCCD1 & VSSD1



```
Writing OpenROAD database to '/tmp/aokiehl/new_tapeout_workspace/openlane/user_project_wrapper_flattened/runs/26_04_30_22_11/20-openroad-generatepdn/user_project_wrapper.odb'...
Writing netlist to '/tmp/aokiehl/new_tapeout_workspace/openlane/user_project_wrapper_flattened/runs/26_04_30_22_11/20-openroad-generatepdn/user_project_wrapper.nl.v'...
Writing powered netlist to '/tmp/aokiehl/new_tapeout_workspace/openlane/user_project_wrapper_flattened/runs/26_04_30_22_11/20-openroad-generatepdn/user_project_wrapper.pnl.v'...
Writing layout to '/tmp/aokiehl/new_tapeout_workspace/openlane/user_project_wrapper_flattened/runs/26_04_30_22_11/20-openroad-generatepdn/user_project_wrapper.def'...
Writing timing constraints to '/tmp/aokiehl/new_tapeout_workspace/openlane/user_project_wrapper_flattened/runs/26_04_30_22_11/20-openroad-generatepdn/user_project_wrapper.sdc'...
[INFO PSM-0040] All shapes on net vccd1 are connected.
[INFO PSM-0040] All shapes on net vccd2 are connected.
[INFO PSM-0040] All shapes on net vdda1 are connected.
[INFO PSM-0040] All shapes on net vdda2 are connected.
[INFO PSM-0040] All shapes on net vssd1 are connected.
[INFO PSM-0040] All shapes on net vssd2 are connected.
[INFO PSM-0040] All shapes on net vssa1 are connected.
[INFO PSM-0040] All shapes on net vssa2 are connected.
[02:13:00] VERBOSE Running 'Odb.RemovePDNObstructions' at 'user_project_wrapper_flattened/runs/26_04_30_22_11/21-odb-removepdn-obstructions'... step.py:1146
[02:13:00] INFO 'PDN_OBSTRUCTIONS' is not defined. Skipping 'Odb.RemovePDNObstructions'... odb.py:568
[02:13:00] VERBOSE Running 'Odb.AddRoutingObstructions' at 'user_project_wrapper_flattened/runs/26_04_30_22_11/22-odb-addrouting-obstructions'... step.py:1146
[02:13:00] INFO 'ROUTING_OBSTRUCTIONS' is not defined. Skipping 'Odb.AddRoutingObstructions'... odb.py:568
[02:13:00] VERBOSE Running 'OpenROAD.GlobalPlacementSkipIO' at 'user_project_wrapper_flattened/runs/26_04_30_22_11/23-openroad-globalplacementskipio'... step.py:1146
[02:13:00] INFO I/O pins were loaded from /tmp/aokiehl/new_tapeout_workspace/openlane/user_project_wrapper/fixd_dont_change/user_project_wrapper.def. Skipping the first global placement iteration... openroad.py:1330
[02:13:00] VERBOSE Running 'OpenROAD.IOPlacement' at 'user_project_wrapper_flattened/runs/26_04_30_22_11/24-openroad-ioplacement'... step.py:1146
[02:13:00] INFO I/O pins were loaded from /tmp/aokiehl/new_tapeout_workspace/openlane/user_project_wrapper/fixd_dont_change/user_project_wrapper.def. Skipping OpenROAD.IOPlacement... openroad.py:1087
[02:13:00] VERBOSE Running 'Odb.CustomIOPlacement' at 'user_project_wrapper_flattened/runs/26_04_30_22_11/25-odb-customioplacement'... step.py:1146
[02:13:00] INFO No custom floorplan file configured, skipping... odb.py:718
[02:13:00] VERBOSE Running 'Odb.ApplyDEFTemplate' at 'user_project_wrapper_flattened/runs/26_04_30_22_11/26-odb-applydeftemplate'... step.py:1146
[02:13:00] VERBOSE Logging subprocess to 'user_project_wrapper_flattened/runs/26_04_30_22_11/26-odb-applydeftemplate/odb-applydeftemplate.log'... step.py:1342
Classic - Stage 26 - Apply DEF Template 25/78 0:01:22
```

Corner/Group	Hold Worst Slack	Reg to Reg Paths	Hold TNS	Hold Vio Count	of which reg to reg	Setup Worst Slack	Reg to Reg Paths	Setup TNS	Setup Vio Count	of which reg to reg	Max Cap Violati...	Max Slew Violati...
Overall	0.1086	0.1086	0.0000	0	0	-0.0158	1.2125	-0.0158	1	0	0	0
nom_tt_025C_1v80	0.3172	0.3172	0.0000	0	0	2.1407	5.6132	0.0000	0	0	0	0
nom_ss_100C_1v60	0.8598	0.8598	0.0000	0	0	0.0742	1.3712	0.0000	0	0	0	0
nom_ff_n40C_1v95	0.1118	0.1118	0.0000	0	0	2.9807	7.2505	0.0000	0	0	0	0
min_tt_025C_1v80	0.3100	0.3100	0.0000	0	0	2.2111	5.7090	0.0000	0	0	0	0
min_ss_100C_1v60	0.8504	0.8504	0.0000	0	0	0.1738	1.5442	0.0000	0	0	0	0
min_ff_n40C_1v95	0.1086	0.1086	0.0000	0	0	3.0313	7.3111	0.0000	0	0	0	0
max_tt_025C_1v80	0.3261	0.3261	0.0000	0	0	2.0759	5.5234	0.0000	0	0	0	0
max_ss_100C_1v60	0.8755	0.8755	0.0000	0	0	-0.0158	1.2125	-0.0158	1	0	0	0
max_ff_n40C_1v95	0.1143	0.1143	0.0000	0	0	2.9288	7.1976	0.0000	0	0	0	0